📊 Trader Behavior Analysis Using Market Sentiment

🔹 Methodology

This project analyzes trader behavior by combining trade-level data with market sentiment indicators (Fear & Greed Index). The workflow begins with data preprocessing, where timestamps are converted and datasets are merged on date. Missing values are handled, and new features such as trade direction (is_long, is_short), profitability (is_win), and rolling statistics (PnL mean, win rate, volatility) are created.

Feature engineering plays a key role, incorporating lag variables (pnl_lag1, pnl_lag2) and rolling metrics to capture short-term trading behavior.

A classification model is built to predict next-day profitability, achieving high accuracy (~98%). In parallel, KMeans clustering is applied to segment traders into behavioral groups based on features like trade size, volatility, and win rate.

Finally, results are visualized through a Streamlit dashboard for interactive exploration.

🔹 Key Insights

 # Insight:

- Infrequent traders earn higher PnL than frequent traders, especially during Fear
- Frequent traders have more stable performance but lower profits
- High-size traders perform well in Fear but perform poorly (even losses) in Greed
- Low-size traders maintain consistent profits across all sentiments
- Low-size traders also have higher win rates
- Inconsistent traders show very high profits during Fear but losses during Greed
- Consistent traders maintain steady but lower profits across all sentiments.
- Traders who take controlled risks (low size, selective trading)
- tend to perform more consistently across market conditions.

🔹 Strategy Recommendations

Strategy 1: Trade More During Fear, Be Careful During Greed

- Trade more when market is in **Fear**
- Trade less and be careful during **Greed**

Strategy 2: Use Small Trade Sizes

- Avoid putting too much money in one trade
- Use smaller and controlled trade sizes

Strategy 3: Don't Trade Too Frequently

- Take fewer but **better quality trades**
- Avoid trading again and again without reason

Strategy 4: Change Strategy Based on Market Condition

- During Fear → take opportunities (more active)
- During Greed → reduce risk (be selective)